

Vision-Generative Assurance for Spacecraft Rendezvous and Proximity Operations

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Abstract

Autonomous rendezvous and proximity operations require perception systems that can interpret visually ambiguous spacecraft scenes while remaining bounded by deterministic mission safety rules. This paper presents a preliminary vision-generative architecture for spacecraft rendezvous and proximity operations in which visual observations are converted into structured scene state estimates, mission constraints are bound into machine-checkable rules, and a generative mission procedure model proposes traceable step-by-step actions subject to deterministic safety verification. Reinforcement learning is not used as a direct flight controller; instead, it is limited to offline simulation-based plan evaluation, robustness testing, and procedure repair. As an initial perception foundation, a multi-task spacecraft vision model was trained on the ESA SPARK 2026 Stream 1 dataset for classification, single-object bounding box localization, spacecraft body segmentation, and solar panel segmentation. A supervised EfficientNet-B0 multi-head baseline achieved 0.9843 classification accuracy, 0.9101 bounding box intersection-over-union, 0.7558 body mask intersection-over-union, and 0.2151 solar panel mask intersection-over-union on the validation set. A second experiment used pair-based view-transformation pretraining inspired by correspondence learning before fine-tuning on the same multi-task objective. The pretraining stage slightly improved classification but did not improve the overall multi-task validation score, indicating that same-image transformed-pair pretraining was insufficiently aligned with thin-structure segmentation. The results motivate a layered RPO autonomy architecture in which generative planning is made useful through rule traceability, explicit uncertainty gating, offline reinforcement learning verification, and an independent deterministic abort authority.

Nomenclature

$\mathcal{I}$	input image or image sequence
$\hat{c}$	predicted spacecraft class

$\hat{b}$	predicted bounding box
$\hat{M}_b$	predicted spacecraft body mask
$\hat{M}_p$	predicted solar panel mask
S	structured scene state
U	uncertainty vector associated with scene state estimates
$\mathcal{R}$	mission rule set
$\mathcal{C}$	machine-checkable constraint set
Π	generated mission procedure
σ	safety shield decision
J	robustness or safety cost used in offline reinforcement learning verification
θ	learned model parameters
RPO	rendezvous and proximity operations
VLM	vision-language model
MLLM	multimodal large language model
RL	reinforcement learning
KOZ	keep-out zone
GNC	guidance, navigation, and control
IoU	intersection-over-union
SSL	self-supervised learning

1 Introduction

Rendezvous and proximity operations (RPO) are among the most safety-critical phases of spacecraft operations. During close approach, servicing, inspection, docking, or formation-flying activities, the vehicle must reason about relative geometry, target visibility, sensor uncertainty, keep-out zones, closing rates, and mission-specific constraints. These requirements make RPO a natural candidate for advanced autonomy, but they also make unconstrained learned decision-making difficult to justify. A learned system may improve perception and planning flexibility, but flight rules, abort logic, and hard safety constraints must remain explicit, auditable, and deterministic.

This paper proposes a vision-generative architecture for RPO in which learned perception and generative planning are surrounded by a strict safety structure. The central idea is that a visual model should not directly command the vehicle. Instead, the visual model converts image data into a structured scene state with uncertainty. A rule-binding layer converts mission guidelines into machine-checkable constraints. A generative mission procedure model then proposes step-by-step procedures with hold points, constraint references, and abort logic. A deterministic safety shield verifies each generated procedure before execution. Reinforcement learning is used only offline, in simulation, to stress-test and repair procedures under uncertainty. It does not command actuators in flight.

The immediate technical focus of this work is the perception foundation needed to support such a system. The ESA SPARK 2026 Stream 1 dataset was used as a synthetic space-

craft perception benchmark for classification, bounding box localization, and segmentation of spacecraft body and solar panels. A series of notebooks was developed to train a supervised multi-task baseline, perform pair-based view-style pretraining, fine-tune the pretrained encoder, and generate final test-set submissions. These experiments serve as a preliminary demonstration of how a spacecraft visual perception module can be developed, evaluated, and prepared for integration into a broader RPO autonomy architecture.

The contributions of this paper are as follows:

1. A layered vision-generative RPO autonomy architecture that separates perception, mission rule binding, generative procedure construction, deterministic safety verification, offline reinforcement learning repair, and monitored execution.
2. A supervised multi-task spacecraft perception pipeline trained on SPARK 2026 Stream 1 for classification, single-object localization, body segmentation, and solar panel segmentation.
3. A pair-based view-transformation pretraining experiment intended to improve correspondence-aware spacecraft representations before supervised fine-tuning.
4. A comparison between the supervised baseline and the pair-pretrained fine-tuned model, showing that the current pretraining task improves image-level recognition only marginally and does not yet improve thin-structure segmentation.

2 Background and Motivation

2.1 RPO Autonomy and the Need for Assurance

RPO autonomy requires both geometric understanding and operational discipline. The vehicle must estimate relative pose, recognize visible structures, maintain safe separation, respect keep-out regions, and react safely to uncertain or degraded sensor conditions. A visual autonomy system that simply maps camera observations to actions is not sufficient for safety-critical use. The system must be able to say no, hold position, retreat, or abort when state estimates become uncertain or when the proposed plan violates flight rules.

The proposed framework therefore treats learned models as bounded components. Perception is learned, but uncertainty is explicit. Procedure generation is generative, but mission rules are injected as constraints rather than learned preferences. Reinforcement learning can search for failure modes and repair weak plans in simulation, but cannot directly control the spacecraft. Final execution authority remains with deterministic safety logic and an independent abort monitor.

2.2 Visual Correspondence and Spacecraft Perception

Visual ambiguity is a central bottleneck in proximity operations. A target spacecraft can appear at different scales, orientations, illumination angles, and partial occlusion states. Solar panels and appendages are especially difficult because they are thin, sometimes dark,

and often affected by aliasing or compression artifacts. Correspondence-aware pretraining is attractive because it encourages a model to learn what remains stable across view changes rather than only memorizing class-specific texture cues.

Recent work in image-goal navigation and 3D vision has shown that cross-view or paired-image pretraining can improve representations for geometry-sensitive downstream tasks. In this work, that idea is adapted in a lightweight way: two aggressively transformed views of the same spacecraft image are used to train an encoder before supervised fine-tuning. This is not a full physical multi-view reconstruction task, but it provides an initial test of whether correspondence-style pretraining improves SPARK perception performance.

3 Vision-Generative RPO Architecture

3.1 System Overview

Figure 1 shows the proposed high-level architecture. The system is organized into runtime and offline assurance paths. The runtime path converts visual observations into scene state estimates and mission procedures, while the offline path improves perception and stress-tests procedures in simulation.

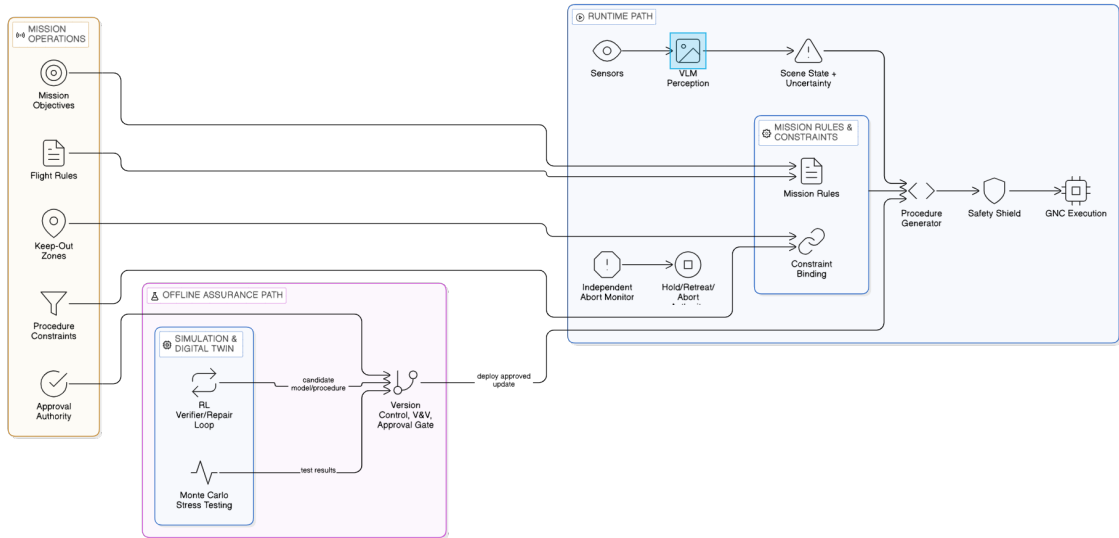


Figure 1: Proposed vision-generative assurance architecture for spacecraft rendezvous and proximity operations. The runtime path converts sensor observations into structured scene state and uncertainty estimates, binds mission rules into machine-checkable constraints, generates traceable procedures, and passes all candidate actions through a deterministic safety shield before GNC execution. The offline assurance path uses simulation, digital twin testing, Monte Carlo stress testing, and reinforcement-learning-based verifier/repair loops to evaluate candidate models and procedures. Offline outputs can affect runtime only through a version-controlled verification, validation, and approval gate; no online learning or direct reinforcement-learning-to-actuator path is permitted.

3.2 Perception Layer

The perception layer receives one or more spacecraft images and outputs a structured scene state:

$$(S, U) = f_{\theta}(\mathcal{I}), \quad (1)$$

where S contains estimated semantic and geometric state variables and U contains uncertainty estimates. In the current experimental implementation, the model outputs a spacecraft class $\hat{c}$, bounding box $\hat{b}$, body mask $\hat{M}_b$, and panel mask $\hat{M}_p$. In a complete RPO system, these outputs would be extended into relative pose, line-of-sight geometry, visible appendage state, range band, approach corridor status, and sensor confidence flags.

3.3 Mission Guideline Binding

The mission guideline binding layer converts human-readable mission constraints into a machine-checkable constraint set:

$$\mathcal{C} = g(\mathcal{R}, S, U), \quad (2)$$

where $\mathcal{R}$ represents flight rules and mission-specific operating constraints. Examples include maximum closure rate, keep-out zone boundaries, minimum sensor confidence thresholds, mandatory hold points, attitude restrictions, and retreat conditions. This layer defines what is allowed, not what action should be taken.

3.4 Generative Procedure Layer

The generative procedure model receives the current scene state and active constraints and proposes a procedure:

$$\Pi = h(S, U, \mathcal{C}). \quad (3)$$

The procedure is not a low-level control command. It is a structured sequence of mission steps, each containing preconditions, action intent, constraints, hold points, expected outcomes, abort triggers, and rule traceability. A representative procedure step is shown below:

Advance to observation hold point. Preconditions: target visible, relative state confidence above threshold, no keep-out zone violation predicted. Constraints: maintain maximum closure rate, remain outside approach exclusion cone, preserve line-of-sight to target body. Hold: stop at waypoint and re-evaluate pose and panel visibility. Abort: if confidence falls below threshold or keep-out margin is violated, command HOLD or RETREAT. Traceability: FR-02, FR-07, SC-04.

3.5 Deterministic Safety Shield

The deterministic safety shield evaluates the generated procedure before execution:

$$\sigma = q(\Pi, S, U, \mathcal{C}), \quad (4)$$

where $\sigma \in \{\text{APPROVE}, \text{HOLD}, \text{RETREAT}, \text{ABORT}, \text{REPAIR}\}$. The shield enforces hard constraints regardless of the generative model output. If a proposed step violates a keep-out boundary, closure-rate constraint, confidence gate, or mandatory hold requirement, it is rejected or replaced by a safe fallback.

3.6 Offline Reinforcement Learning Verification and Repair

Reinforcement learning is used only in the offline assurance loop. Candidate procedures are tested under simulated uncertainty, degraded perception, lighting variation, and relative-state perturbations. The RL verifier attempts to identify weak procedure sequences and propose repairs that reduce expected safety cost:

$$\Pi^* = \arg \min_{\Pi} \mathbb{E} [J(\Pi, \delta S, \delta U, \mathcal{E})], \quad (5)$$

where δS and δU represent state and uncertainty perturbations, and $\mathcal{E}$ represents the simulated environment and mission phase. Repaired plans are still passed through the deterministic shield before being considered valid. No online weight updates or RL policy execution are permitted in flight.

4 SPARK Perception Dataset

The SPARK 2026 Stream 1 dataset contains synthetic spacecraft images organized for multi-task perception. Each image contains one spacecraft rendered under varying lighting, orientation, scale, and background conditions. The labeled training and validation data include class labels, bounding boxes, and segmentation masks. The test set contains 5000 unlabeled images, for which the model must output a detection CSV and one compressed segmentation mask file per image.

The training set contains 60,000 images and the validation set contains 20,000 images across 10 spacecraft classes: Cheops, LisaPathfinder, ObservationSat1, Proba2, Proba3, Proba3ocs, Smart1, Soho, VenusExpress, and XMM Newton. Images are stored under class-specific folders, while masks are stored in a mirrored directory structure. The CSV files provide image names, mask names, class names, and bounding box coordinates.

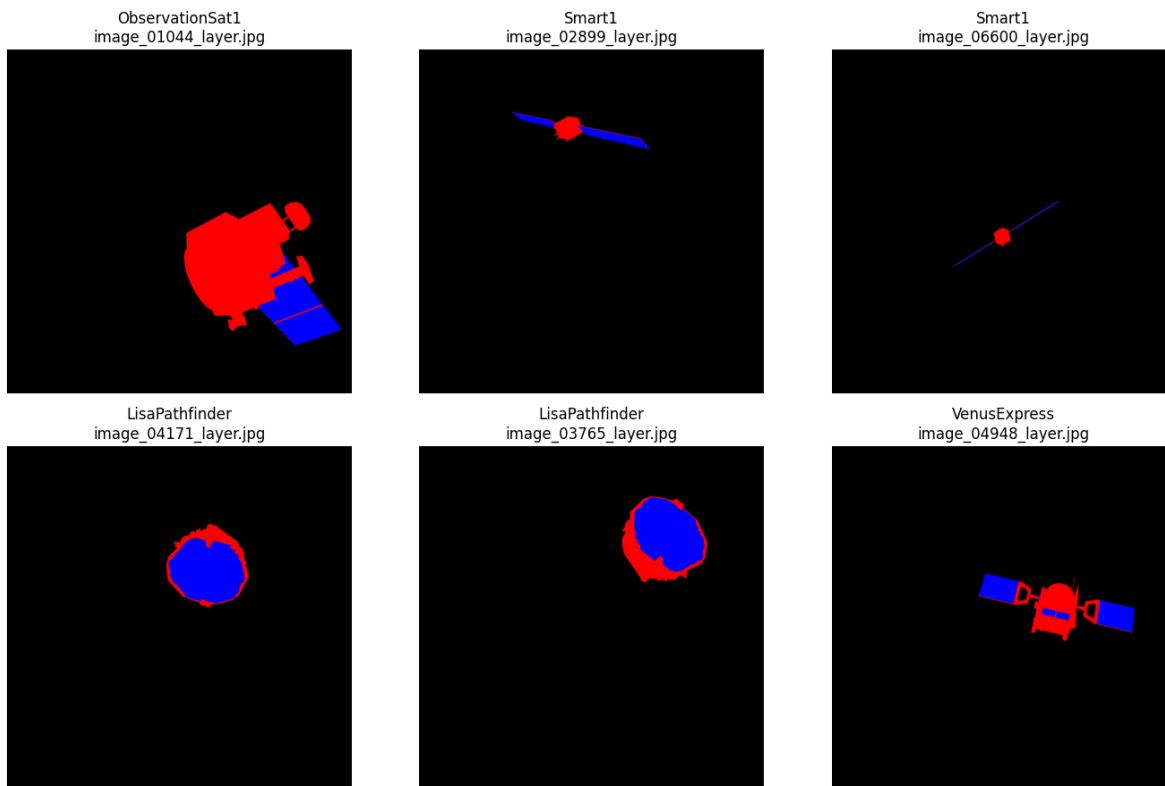


Figure 2: Representative SPARK 2026 Stream 1 segmentation masks across multiple spacecraft classes. Red regions correspond primarily to spacecraft body structure, while blue regions correspond primarily to solar-panel regions. The examples show the variation in spacecraft scale, orientation, and projected geometry.

A challenge encountered during preprocessing was that the segmentation masks were stored as JPEG images with thousands of unique pixel values due to compression artifacts. Exact RGB thresholding was therefore unreliable. The final preprocessing approach converted masks to grayscale, separated foreground from background, and split foreground intensities into body and panel regions. This approach produced usable segmentation supervision and enabled the body segmentation IoU to improve substantially from the initial broken-mask runs.

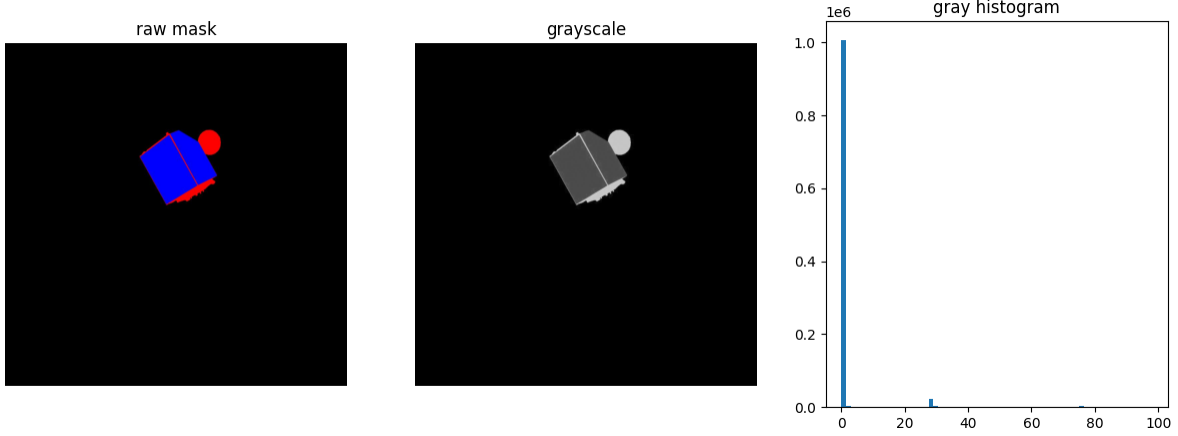


Figure 3: Example of the grayscale-based mask extraction process used to separate spacecraft body and solar-panel regions from JPEG-compressed segmentation masks. This preprocessing step was required because exact RGB thresholding was unreliable due to compression artifacts.

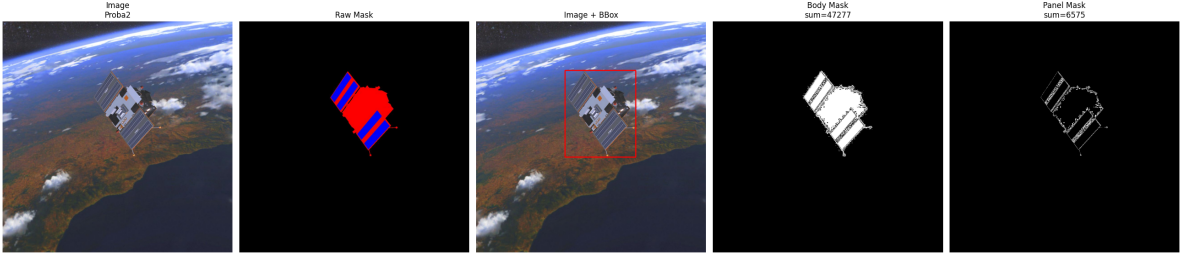


Figure 4: Additional mask extraction example showing the raw mask, bounding box overlay, body target, and panel target used during dataset verification.

5 Modeling and Training Method

5.1 Supervised Multi-Task Baseline

The baseline model uses an EfficientNet-B0 encoder initialized from ImageNet weights. A global pooling layer feeds a classification head and bounding box regression head. The dense encoder output is reduced with a 1×1 convolution and passed through a lightweight transpose-convolution decoder to produce two binary segmentation logits: one for spacecraft body and one for solar panels.

The model outputs are:

$$f_{\theta}(\mathcal{I}) = \left(\hat{c}, \hat{b}, \hat{M}_b, \hat{M}_p \right). \quad (6)$$

The total training loss is:

$$\mathcal{L} = \lambda_c \mathcal{L}_{cls} + \lambda_b \mathcal{L}_{box} + \lambda_{body} \mathcal{L}_{body} + \lambda_{panel} \mathcal{L}_{panel}, \quad (7)$$

where $\mathcal{L}_{cls}$ is cross-entropy loss, $\mathcal{L}_{box}$ is Smooth L1 loss, and the segmentation losses combine binary cross entropy and Dice loss. The solar panel loss was weighted more heavily because panel segmentation was consistently the hardest task.

5.2 Pair-Based View-Transformation Pretraining

The second notebook introduced a pair-pretraining stage. For each source image, two augmented views were generated using aggressive but controlled transforms: affine rotation, scale perturbation, translation, brightness and contrast variation, blur, and coarse dropout. The encoder processed both views with shared weights. A projection head produced normalized embeddings, while auxiliary heads predicted relative rotation, scale, and occlusion bins.

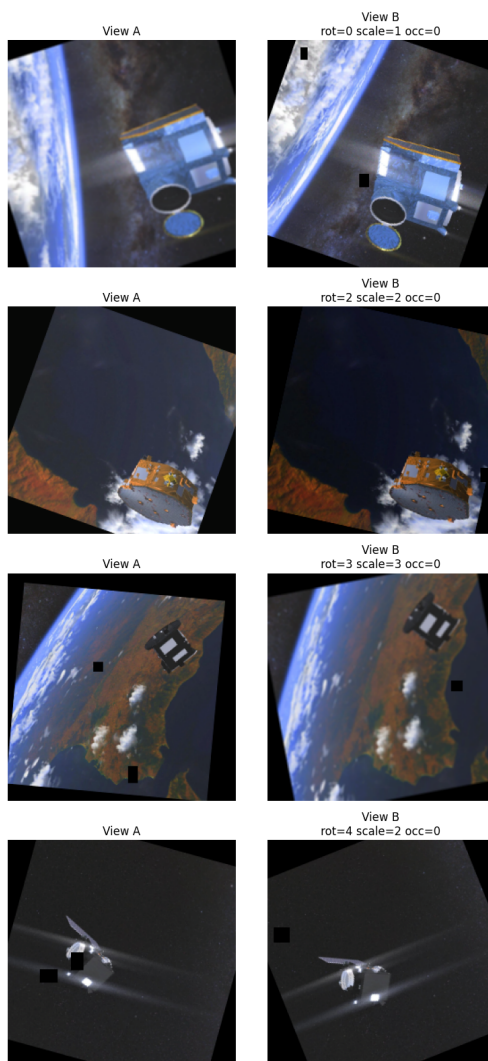


Figure 5: Examples of paired image views used during correspondence-style pretraining. Each pair is generated from the same spacecraft image using independent view-style augmentations, including rotation, scale changes, brightness variation, blur, and partial occlusion.

The pretraining loss was:

$$\mathcal{L}_{pre} = \lambda_m \mathcal{L}_{match} + \lambda_z \mathcal{L}_{contrast} + \lambda_r \mathcal{L}_{rot} + \lambda_s \mathcal{L}_{scale} + \lambda_o \mathcal{L}_{occ}. \quad (8)$$

The intent was to make the encoder more robust to viewpoint-like changes before supervised fine-tuning.

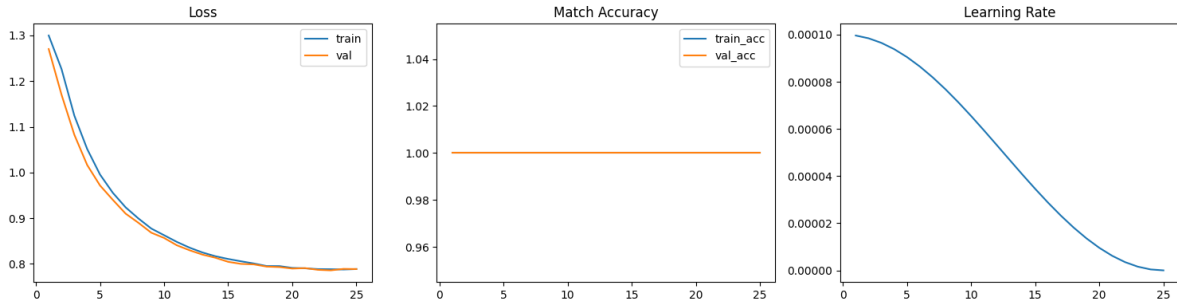


Figure 6: Pair-pretraining learning curves. The loss decreases smoothly while match accuracy remains saturated, indicating that the same-image transformed-pair objective is stable but likely too easy to strongly improve downstream segmentation.

5.3 Fine-Tuning of the Pair-Pretrained Encoder

The third notebook loaded the encoder weights from the pair-pretraining stage into the supervised multi-task architecture. Lower encoder layers were frozen for the first portion of training, then all layers were unfrozen. Differential learning rates were used: a lower rate for the encoder and a higher rate for task heads. This enabled direct comparison between an ImageNet-initialized supervised baseline and a pair-pretrained supervised model.

6 Experimental Results

6.1 Training Configuration

The supervised baseline was trained at 384×384 input resolution using a batch size of 4 on a Kaggle Tesla T4 GPU environment. The best baseline run used an increased solar panel loss weight and trained for 15 epochs. The pair-pretraining stage was trained separately at 256×256 input resolution and saved as an encoder-only checkpoint. The fine-tuning stage then loaded this encoder and trained on the full supervised SPARK task.

6.2 Validation Results

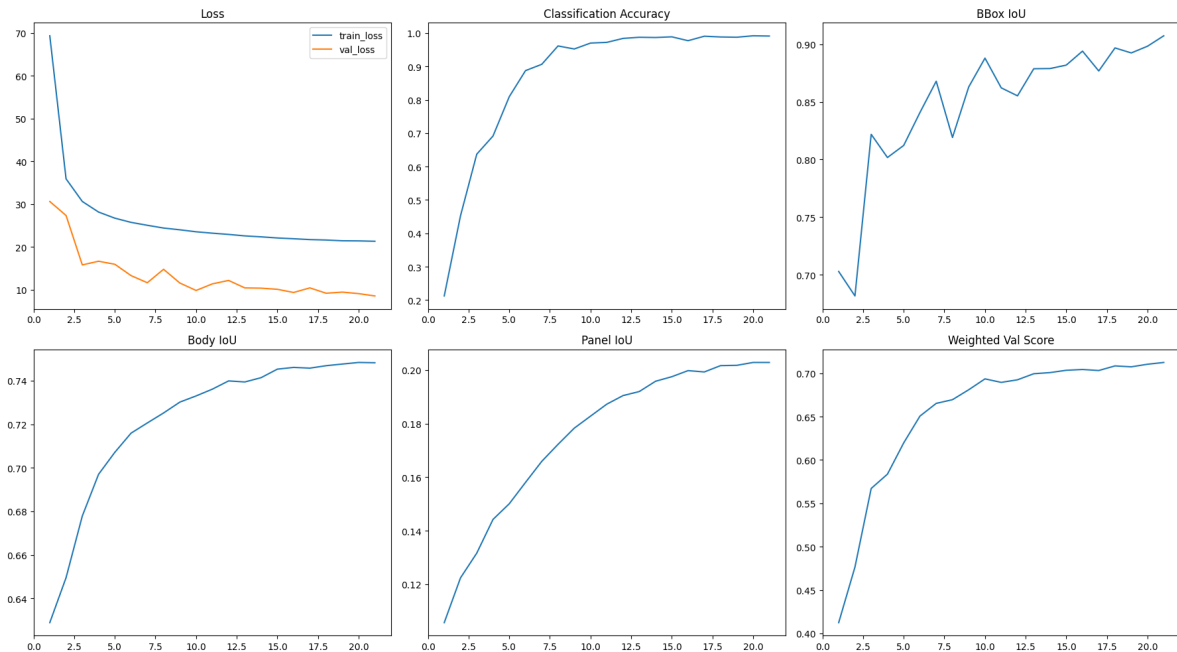


Figure 7: Training and validation trends for the supervised multi-task baseline. Classification accuracy and bounding-box IoU improve rapidly, while body segmentation improves steadily. Solar-panel IoU remains the limiting metric because of thin structures and low pixel occupancy.

Table 2 compares the best supervised baseline and the pair-pretrained fine-tuned model.

Table 2: Validation performance of supervised baseline and pair-pretrained fine-tuned model.

Model	Class Acc.	BBox IoU	Body IoU	Panel IoU	Score
Supervised baseline	0.9843	0.9101	0.7558	0.2151	0.7163
Pair-pretrained fine-tuned	0.9912	0.9074	0.7483	0.2028	0.7124

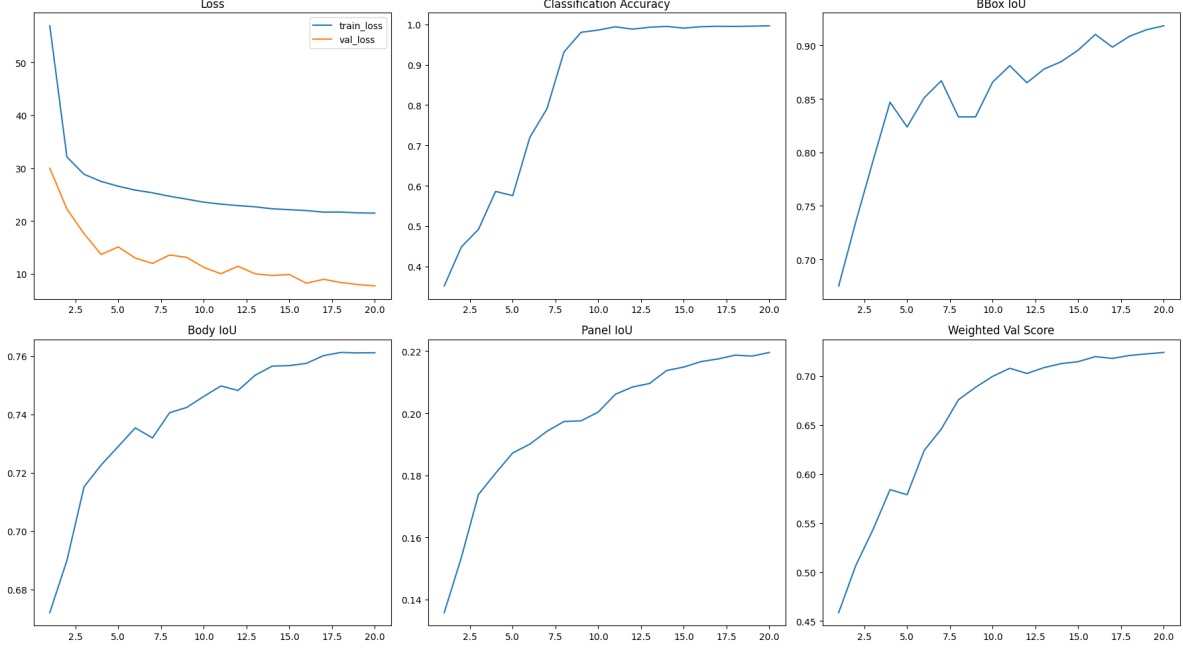


Figure 8: Fine-tuning trends after initializing the supervised multi-task model with the pair-pretrained encoder. The model reaches strong classification and localization performance, but the pair-pretrained initialization does not outperform the best supervised baseline on segmentation-heavy validation metrics.

The supervised baseline achieved the highest overall validation score. The pair-pretrained fine-tuned model improved classification accuracy but slightly reduced localization and segmentation performance. This indicates that the current pair-pretraining objective was not sufficiently aligned with downstream segmentation, especially the thin solar panel target.

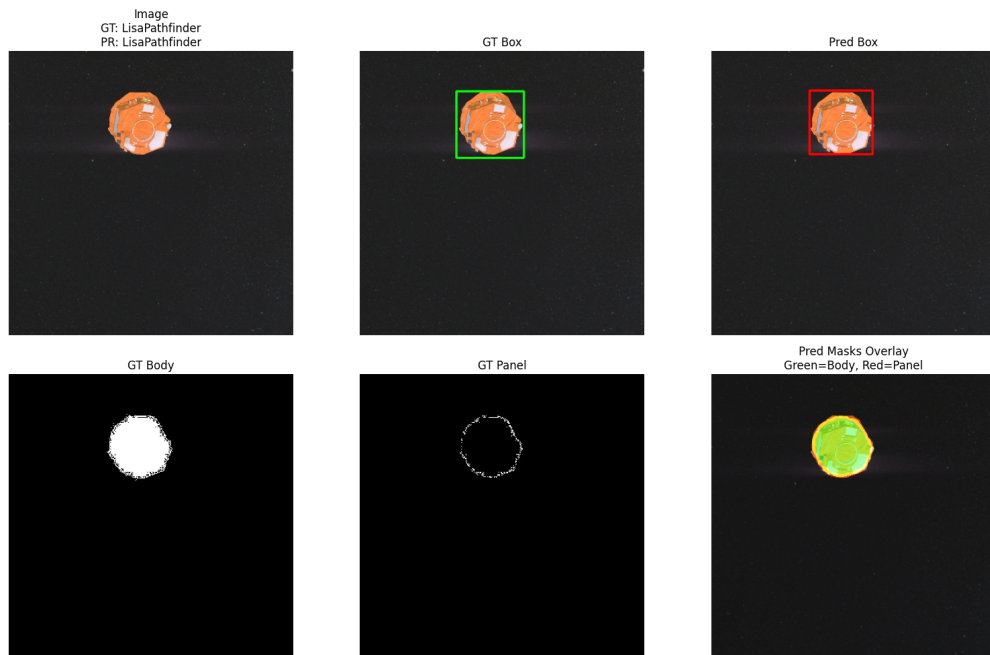


Figure 9: Representative validation prediction from the supervised multi-task baseline. The model correctly predicts spacecraft class and bounding box while producing body and solar-panel segmentation outputs.

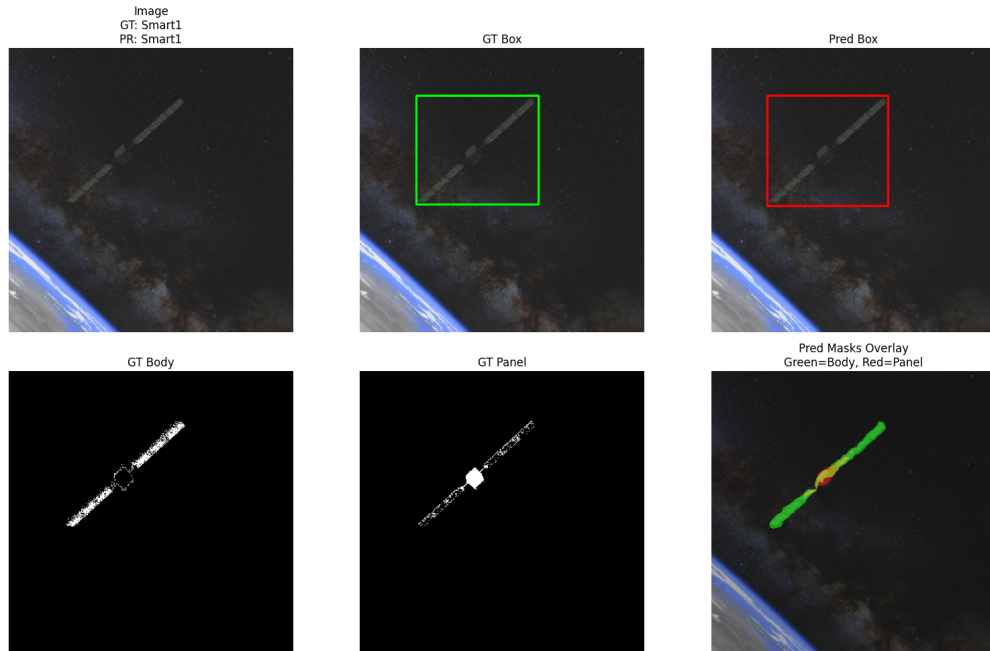


Figure 10: Additional supervised baseline prediction example showing class recognition, bounding-box localization, and segmentation behavior on a validation image.

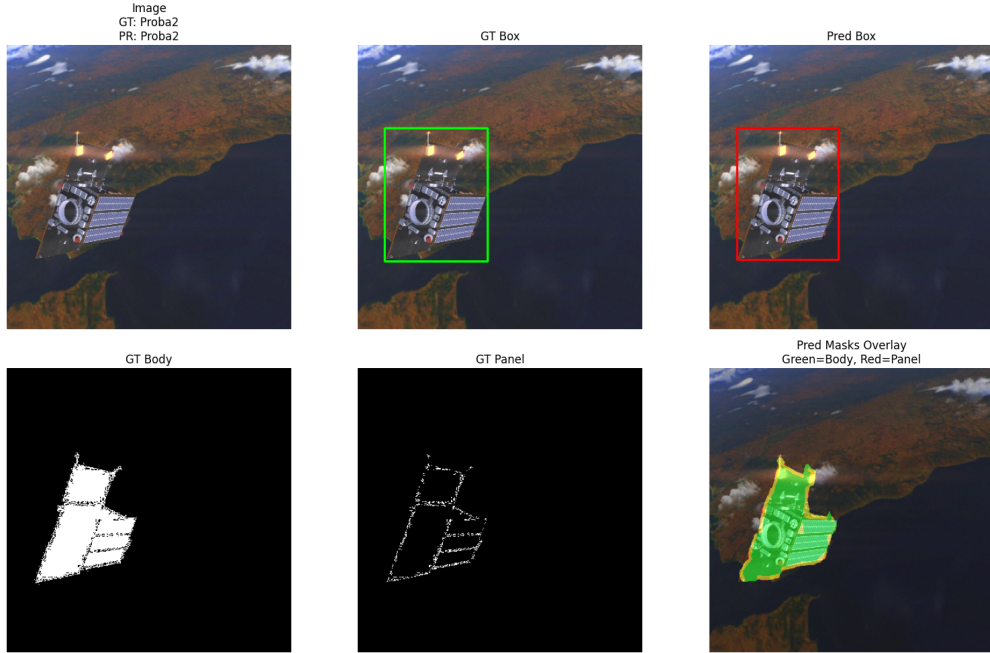


Figure 11: Representative validation prediction after fine-tuning the pair-pretrained encoder on the supervised SPARK multi-task problem. The result shows strong localization and classification, while segmentation remains limited by thin solar-panel structure.

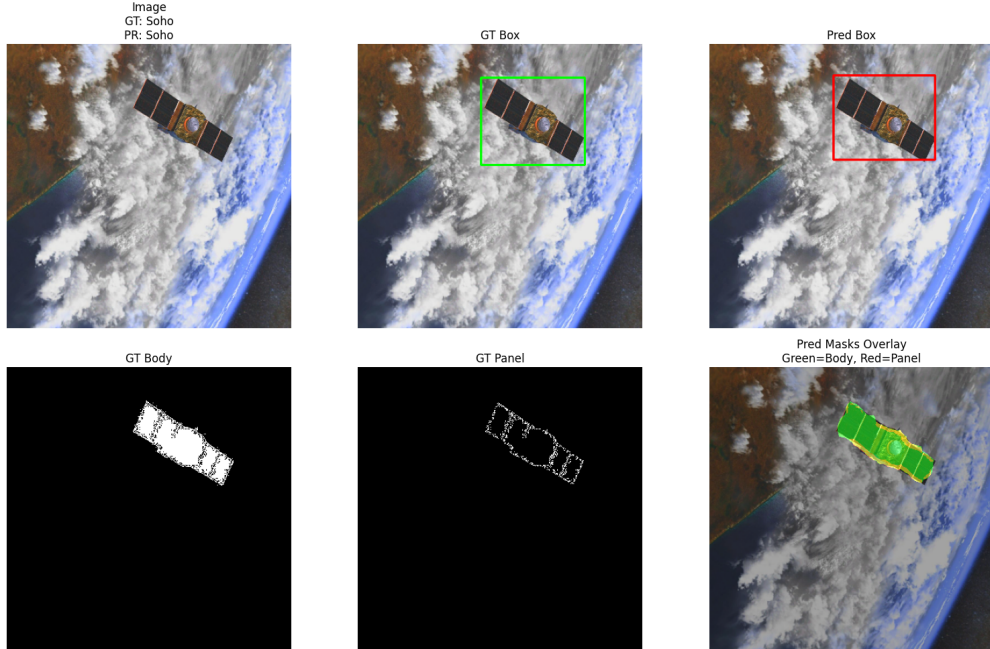


Figure 12: Additional fine-tuned validation prediction showing spacecraft class recognition, bounding-box localization, and body and solar-panel segmentation outputs.

6.3 Test Submission Generation

The fourth notebook loaded the best supervised baseline checkpoint and ran inference on the 5000 unlabeled test images. The inference script generated a `detection.csv` file containing predicted class names and bounding boxes, and one `.npz` file per test image containing a three-channel boolean segmentation mask. Channel 0 stored spacecraft body, channel 1 stored solar panels, and channel 2 was reserved. These files were packaged into `submission.zip` for evaluation.

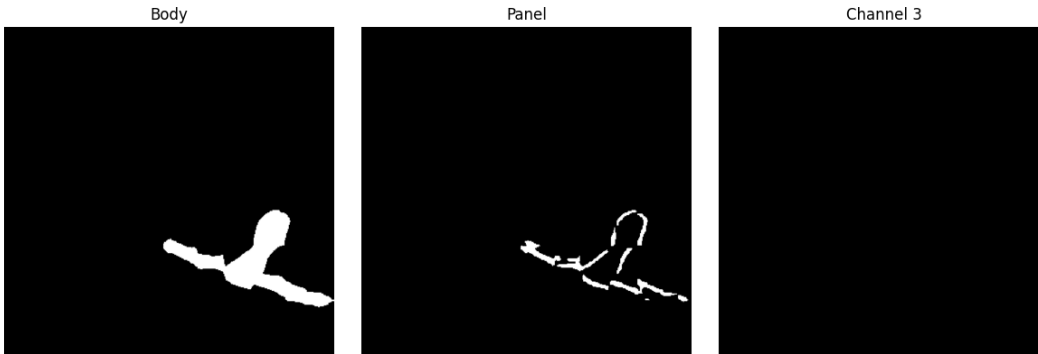


Figure 13: Example of the generated test-set segmentation output format. The submitted mask is stored as a three-channel boolean array, where channel 0 corresponds to spacecraft body, channel 1 corresponds to solar panels, and channel 2 is reserved or unused.

7 Discussion

7.1 Interpretation of Results

The results show that a relatively lightweight shared-backbone multi-task model can learn strong spacecraft classification and bounding box localization from synthetic orbital imagery. Body segmentation also reached a usable level of performance. Solar panel segmentation remained the main bottleneck. This result is expected because panels are thin, occupy fewer pixels than the main body, and are more sensitive to downsampling, compression artifacts, illumination, and ambiguity in the grayscale mask split.

The pair-pretraining experiment did not improve the final score. Although it produced a stable encoder and high pair-match performance, the task was likely too easy. Since both views came from the same original image, the network could learn same-image augmentation invariance without necessarily learning the geometry needed to improve thin-structure segmentation. Future pretraining should use true multi-view synthetic spacecraft pairs, harder negatives, same-class different-instance pairs, or explicit mask and correspondence objectives.

7.2 Relevance to RPO

The perception results are not a complete RPO autonomy solution. They are a first step toward a structured perception layer. In a full RPO system, visual outputs must be con-

verted into mission-relevant quantities such as relative pose, keep-out margin, range band, appendage visibility, and confidence. The current model demonstrates that synthetic spacecraft imagery can support a multi-task visual state estimator, but additional uncertainty calibration and geometry-aware training are needed before the output can support safety-critical procedure generation.

7.3 Assurance Principles

The proposed architecture is intentionally conservative about learned autonomy. The generative procedure model does not modify flight rules. Reinforcement learning does not command the vehicle. The deterministic safety shield remains the final authority before execution, and an independent monitor can trigger HOLD, RETREAT, or ABORT. This hierarchy is essential for any credible use of generative models in proximity operations.

8 Future Work

Future work will focus on both perception improvements and architecture-level assurance.

1. **Improved panel segmentation:** use higher-resolution training, boundary-aware losses, panel oversampling, and a stronger U-Net or FPN-style decoder.
2. **Better correspondence pretraining:** replace same-image transformed pairs with rendered multi-view spacecraft pairs generated in Blender or a spacecraft simulator.
3. **Uncertainty estimation:** add Monte Carlo dropout, ensembles, or evidential outputs to produce confidence estimates for class, box, and masks.
4. **Scene-state conversion:** convert perception outputs into structured RPO state variables such as visible appendages, target bounding geometry, approach corridor status, and sensor confidence flags.
5. **Procedure generation:** develop a constrained MLLM prompt and schema for generating traceable RPO procedures with hold points and abort logic.
6. **Offline RL verification:** build a simulation loop in which RL stress-tests generated procedures under uncertain relative states and proposes repairs that remain subject to deterministic safety verification.

9 Conclusion

This paper presented a preliminary vision-generative architecture for spacecraft rendezvous and proximity operations and developed a perception foundation using the SPARK 2026 Stream 1 dataset. A supervised EfficientNet-B0 multi-task baseline achieved strong validation performance in classification, bounding box localization, and body segmentation, while solar panel segmentation remained the primary technical bottleneck. A pair-based view-transformation pretraining stage was implemented and successfully fine-tuned, but it did

not outperform the supervised baseline overall. These results suggest that future pretraining must be more geometry-aware and more directly aligned with thin-structure segmentation.

The broader contribution is an assurance-centered autonomy concept: learned perception and generative planning can be useful for RPO only when bounded by explicit uncertainty, fixed mission rules, deterministic safety verification, offline-only reinforcement learning repair, and independent abort authority. This structure provides a path toward using vision-generative models in spacecraft proximity operations without surrendering the safety hierarchy required for flight-critical systems.

Acknowledgments

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